

Title

Traffic Sign Recognition System

Introduction

This project uses a Convolutional Neural Network (CNN) for recognizing traffic signs based on the Traffic Sign Dataset Classification dataset by Kaggle.

Dataset

- Name: Traffic Sign Dataset Classification
- Number of Classes: 58
- Image Format: RGB Images

Methods

- Image preprocessing
- Data augmentation
- CNN architecture
- Transfer learning approach (MobileNetV2)
- Accuracy & loss analysis
- Confusion matrix
- Classification report

Technology

- Programming Language: Python
- Deep Learning Libraries: TensorFlow/Keras
- Mathematics Library: NumPy
- Visualization Libraries: Matplotlib
- Evaluation Library: Scikit-learn

Output

The CNN model was trained for 5 epochs and its performance was measured in terms of accuracy, loss, confusion matrix, and classification report.

Steps to Run

1. Get the dataset from Kaggle website.
2. Open the notebook in Kaggle.

3. Execute all cells.

4. Visualize output results.